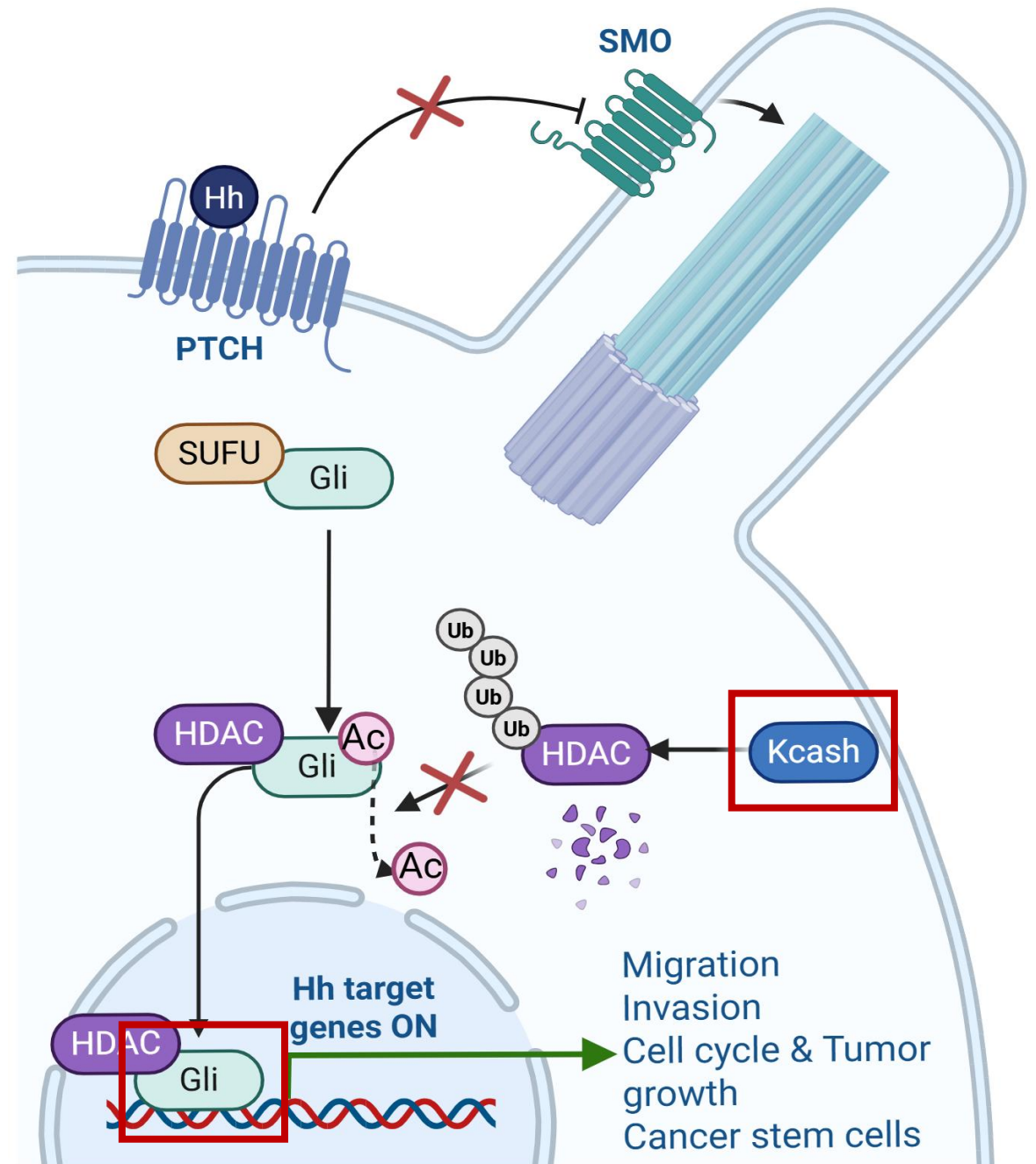


Sonic-Hedgehog (Shh) pathway

- Shh pathway plays a crucial role in embryonic development, tissue differentiation and stem cell maintenance
- Altered Hh signaling is involved in medulloblastoma and basal cell carcinoma; growing evidence also links its deregulation to colorectal cancer (CRC)
- Gli proteins, the final Hh effectors, are the most extensively studied factors
- the KCASH protein family, particularly **KCASH2/KCTD21**, a suppressor of Hh/Gli signaling, has gained attention for its role as tumor suppressor gene



Insights into the role of KCASH2 in colorectal cancer and its microRNA-mediated regulation

This proposal is aimed at better understanding the molecular mechanisms of Hedgehog (Hh) dependent tumorigenesis in CRC and the role of genetic and epigenetic factors (microRNA, miRNA) involved in the modulation of the Hh pathway in this context, with particular reference to the role played by the KCASH2 oncosuppressor.

Among the experiments/analyses performed:

- In vivo analysis (CRC development in KCASH2^{WT} and KCASH2^{KO} mice): KCASH2 knockout mice were used in chemically induced colorectal carcinogenesis protocols in order to evaluate the role of this suppressor in tumor development and Hedgehog pathway activity.
- In silico analysis of KCASH2 expression in CRC patients for the identification of “KCASH2-related” CRC subgroup: investigation of KCASH2 expression levels in a large series of CRC patients (Colorectal Adenocarcinoma TCGA Pan-Cancer Atlas dataset) and evaluation of potential associations between KCASH2 expression and key clinical-pathological/molecular characteristics



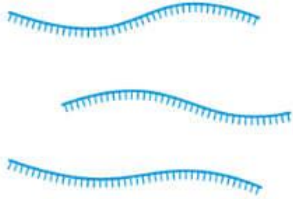
Histopathological analyses and **RNA sequencing of tumor and adjacent normal tissues (~30 samples)**



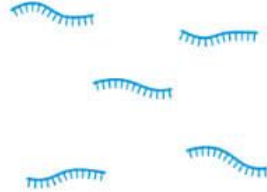
Analysis of clinical, pathological, genomic, transcriptomic, **methylation** and survival data in 592 CRC samples

RNA Sequencing

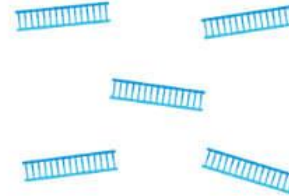
① Isolate RNA from samples



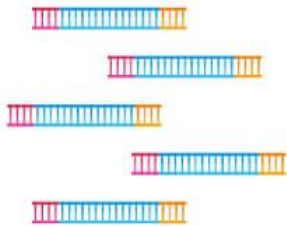
② Fragment RNA into short segments



③ Convert RNA fragments into cDNA



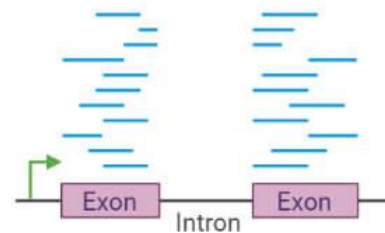
④ Ligate sequencing adapters and amplify



⑤ Perform NGS sequencing

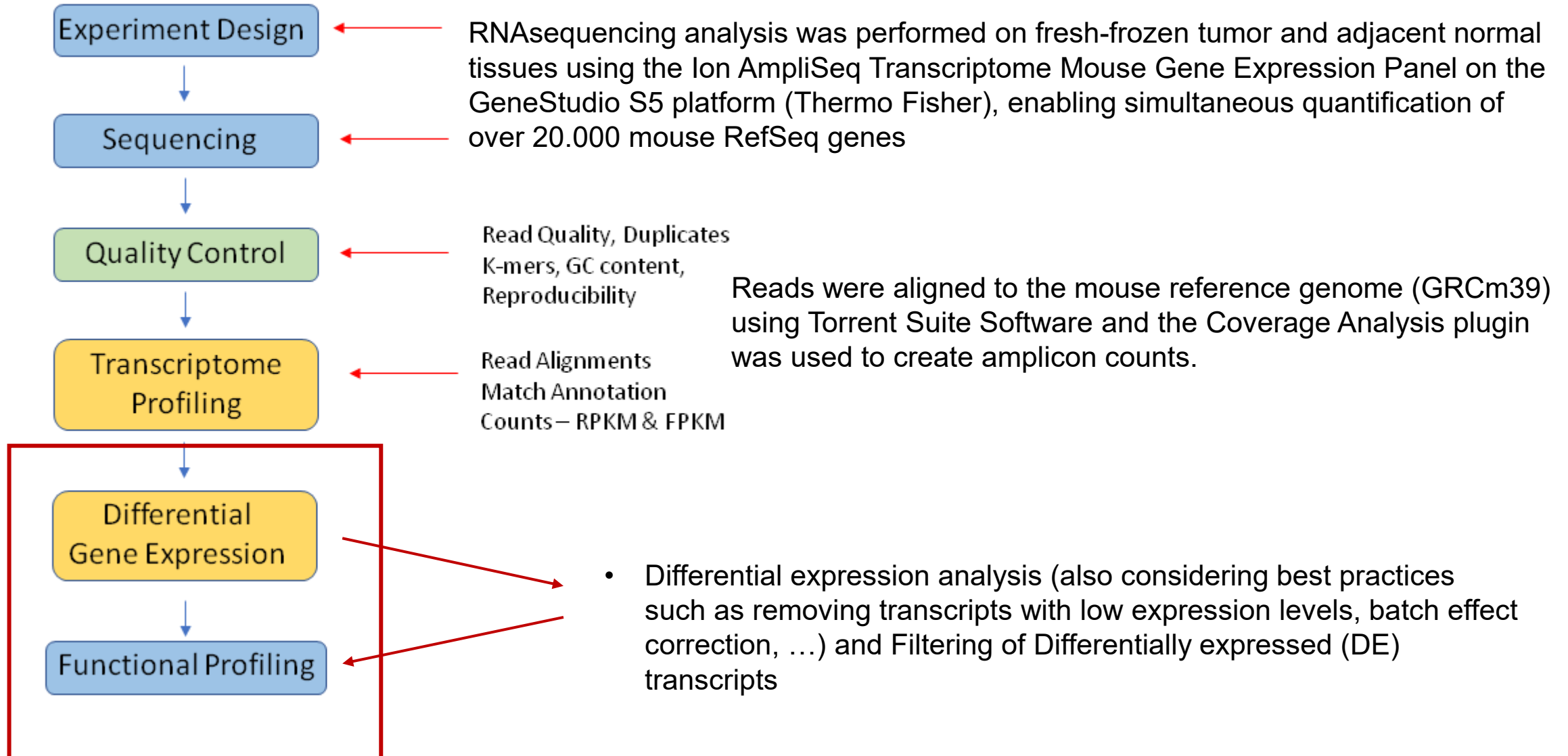


⑥ Map sequencing reads to the transcriptome/genome

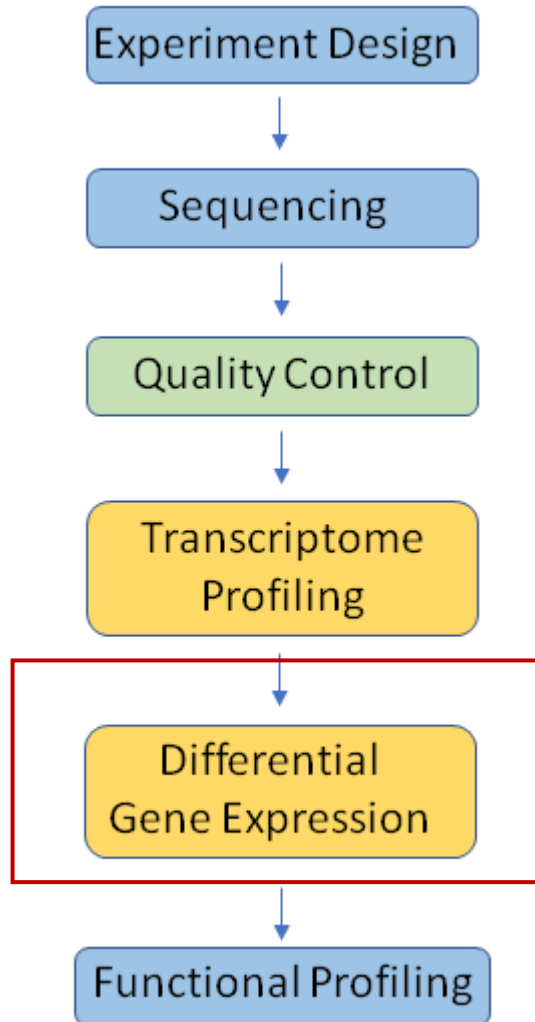


- RNA sequencing (RNA-seq) is a high-throughput technique used to analyze the complete set of RNA transcripts (the transcriptome) in a biological sample.
- It works by converting RNA into complementary DNA (cDNA), which is then sequenced using next-generation sequencing technologies. The resulting reads are aligned to a reference genome to quantify gene expression levels.
- RNA-seq is used to identify which genes are expressed, evaluate differences in gene expression between conditions (e.g., healthy vs diseased), and detect novel transcripts, splice variants, or gene fusions

RNA-Seq Analysis Workflow



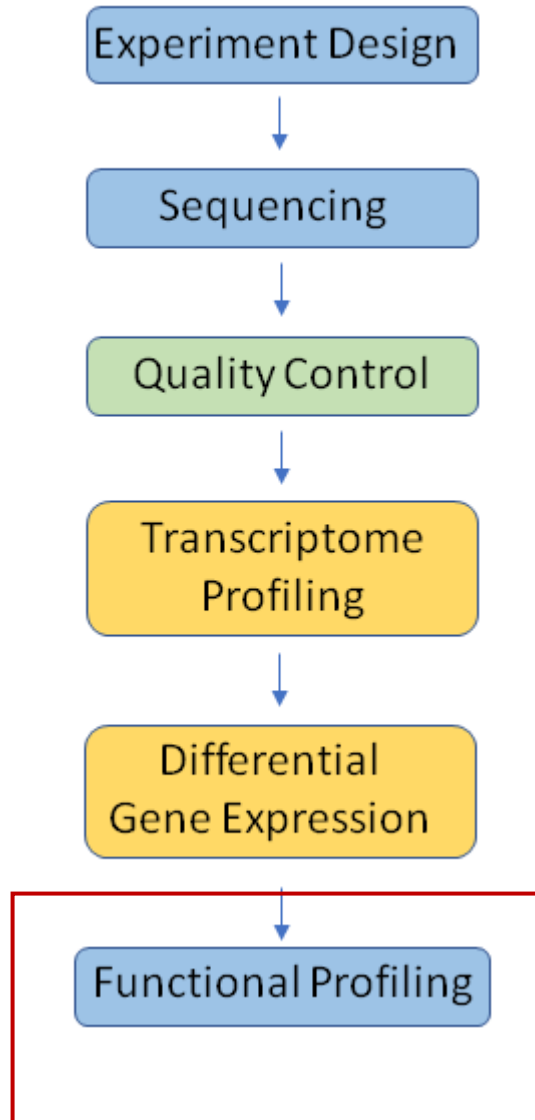
RNA-Seq Analysis Workflow



- **DESeq2 and edgeR** are two widely used tools for performing differential gene expression analysis. These tools compare the read counts between different conditions or groups and identify genes that are significantly differentially expressed.
- **Filtering of Differentially expressed (DE) transcripts** is generally based on a log2 fold change ≤ -1 (down-regulated genes) or ≥ 1 (up-regulated genes) and an adjusted p-value < 0.05 .

<https://pubmed.ncbi.nlm.nih.gov/33926528/>

RNA-Seq Analysis Workflow



Gene Ontology (GO) and pathway analysis of differentially expressed (DE) genes are bioinformatic approaches used to interpret large gene lists by identifying their biological meaning.

GO analysis classifies DE genes into standardized categories, describing their associated biological processes (e.g., cell proliferation), molecular functions (e.g., enzyme activity), and cellular components (e.g., nucleus).

Pathway analysis, maps DE genes onto known biological pathways (such as signaling or metabolic pathways) to identify which cellular processes are significantly affected under specific conditions.

- **Visualization Tools:** Tools like R (with ggplot2 and other visualization packages), Python (with Matplotlib and Seaborn), and interactive tools like Shiny can be employed to generate informative plots and visualizations to aid in interpretation.
- **Database Resources:** databases such as Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG), can help you functionally annotate and interpret differentially expressed genes.

<https://pubmed.ncbi.nlm.nih.gov/26100218/>

Project 1

Study series: 28 samples

- 14 tumors (7 KCASH2-ko, 6 KCASH2-wt)
- 15 adjacent normal tissues (8 KCASH2-ko, 7 KCASH2-wt)

Input files:

- RNAseq raw counts
- Meta data (sample information)

Differential expression analysis (4 analyses):

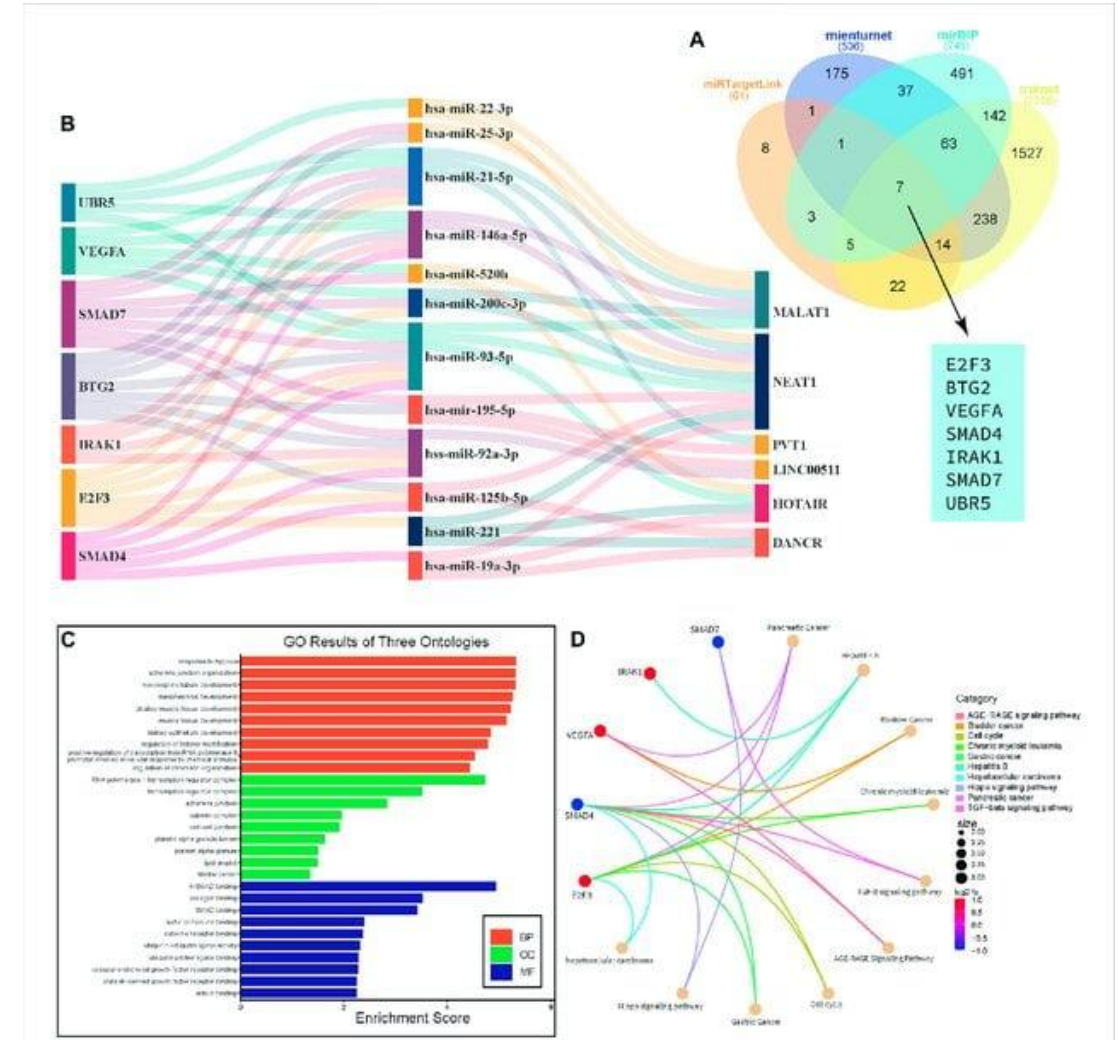
T_ko (n=7) vs T_wt (n=6)

N_ko (n=8) vs N_wt (n=7)

T_ko (n=7) vs N_ko (n=7) (paired)

T_wt (n=6) vs N_wt (n=6) (paired)

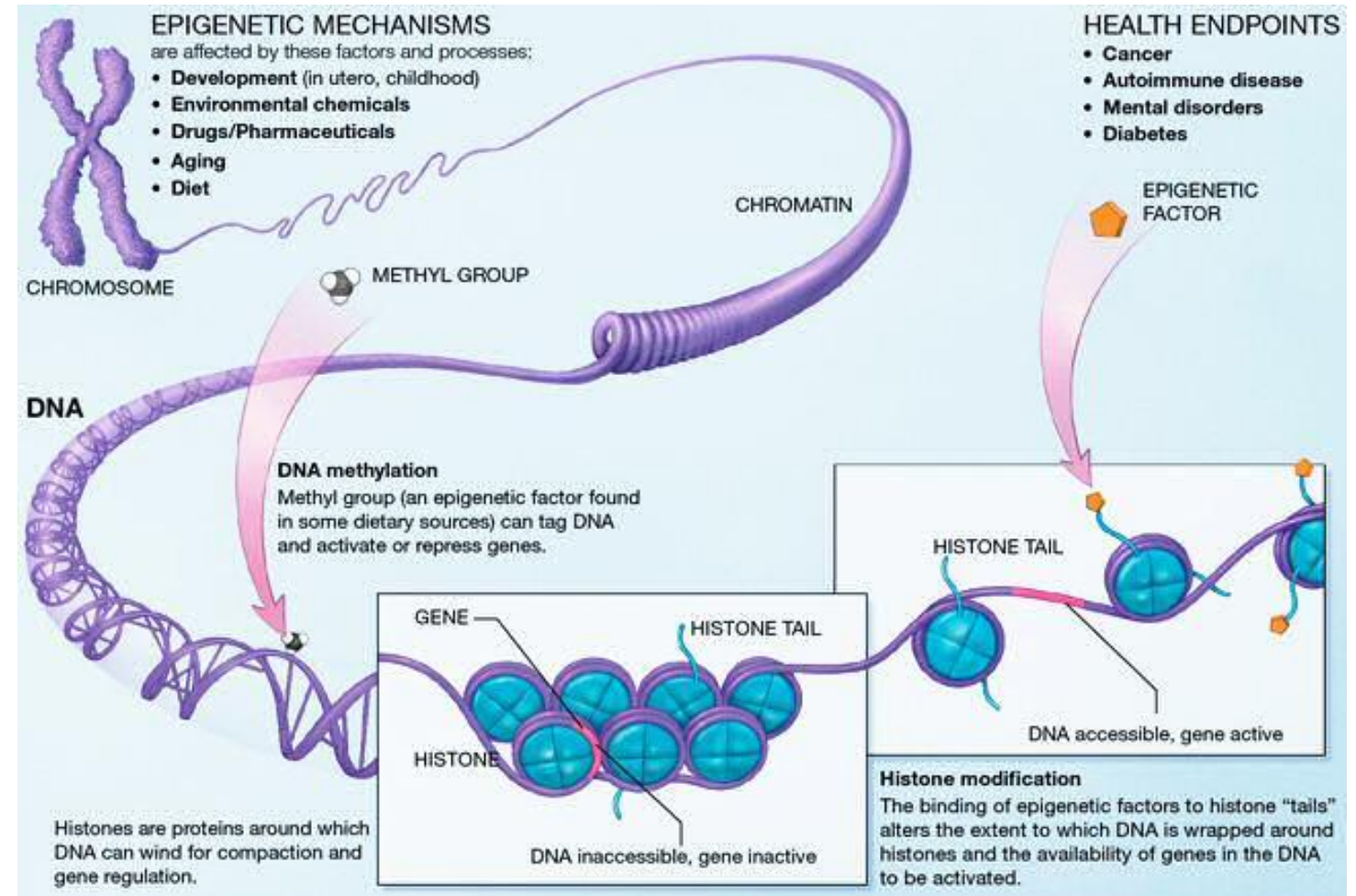
GO and pathways analysis



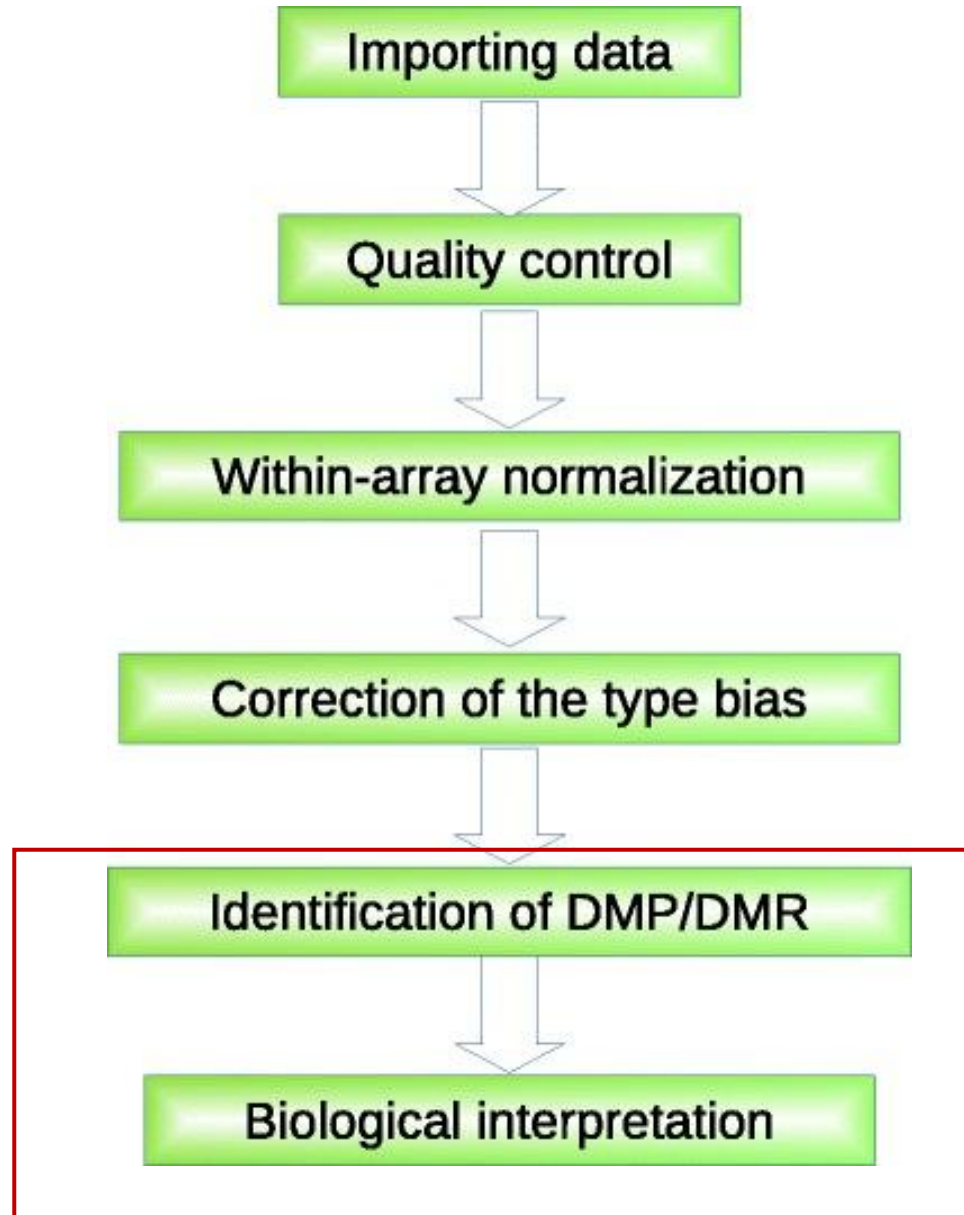
DNA methylation is a chemical modification where a methyl group ($-CH_3$) is added to DNA, usually at cytosines in CpG sites. It does not change the DNA sequence, but it can regulate gene expression, often reducing or silencing it.

Array-based methylation analysis uses microarrays with probes designed to distinguish between methylated and unmethylated DNA at many CpG sites. After a chemical treatment (e.g., bisulfite conversion) that alters unmethylated cytosines, the hybridization signal is evaluated and a “**beta value**” is computed for each site, representing the fraction of methylation.

This approach is used to identify epigenetic differences between samples (e.g., healthy vs diseased), highlighting changes in gene regulation associated with development, environment, or disease.



DNA-methylation Analysis Workflow



- DMPs: Differentially Methylated Positions (individual CpG sites)
- DMRs: Differentially Methylated Regions (clusters of adjacent CpGs showing consistent, biologically significant methylation patterns)

In our data: **differentially methylated genes - DMGs**
(Gene-level methylation differences associated with CRC subgroups)

Expected outputs

- Table of genes:
 - logFC (methylation difference)
 - adjusted p-value
- Heatmap of top genes
- Volcano plot
-

Beta-values are a measure used in DNA methylation array analysis to quantify the level of methylation at a specific CpG site.

They range from 0 to 1, where:

0 indicates no methylation (completely unmethylated)

1 indicates full methylation (completely methylated)

Each beta-value is calculated as the ratio of the methylated signal to the total signal (methylated + unmethylated), reflecting the proportion of DNA molecules that are methylated at that site.

Example of critical steps to consider:

- Many standard pipelines transform beta-values into M-values ($\log_2$ ratio of methylated vs unmethylated signal) before applying linear models, because M-values have more stable statistical properties.
- confounding factors (age, sex,...)

► BMC Bioinformatics. 2018 Apr 11;19(Suppl 5):115. doi: [10.1186/s12859-018-2096-3](https://doi.org/10.1186/s12859-018-2096-3) 

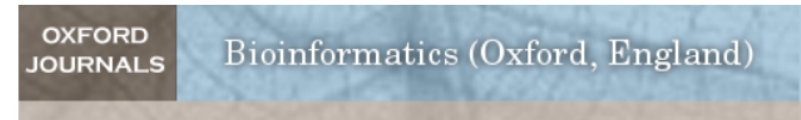
A framework for analyzing DNA methylation data from Illumina Infinium HumanMethylation450 BeadChip

[Zhenxing Wang](#)¹, [XiaoLiang Wu](#)¹, [Yadong Wang](#)¹, 

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PMCID: PMC5907140 PMID: [29671397](#)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5907140/>



► Bioinformatics. 2014 Jan 28;30(10):1363–1369. doi: [10.1093/bioinformatics/btu049](https://doi.org/10.1093/bioinformatics/btu049) 

Minfi: a flexible and comprehensive Bioconductor package for the analysis of Infinium DNA methylation microarrays

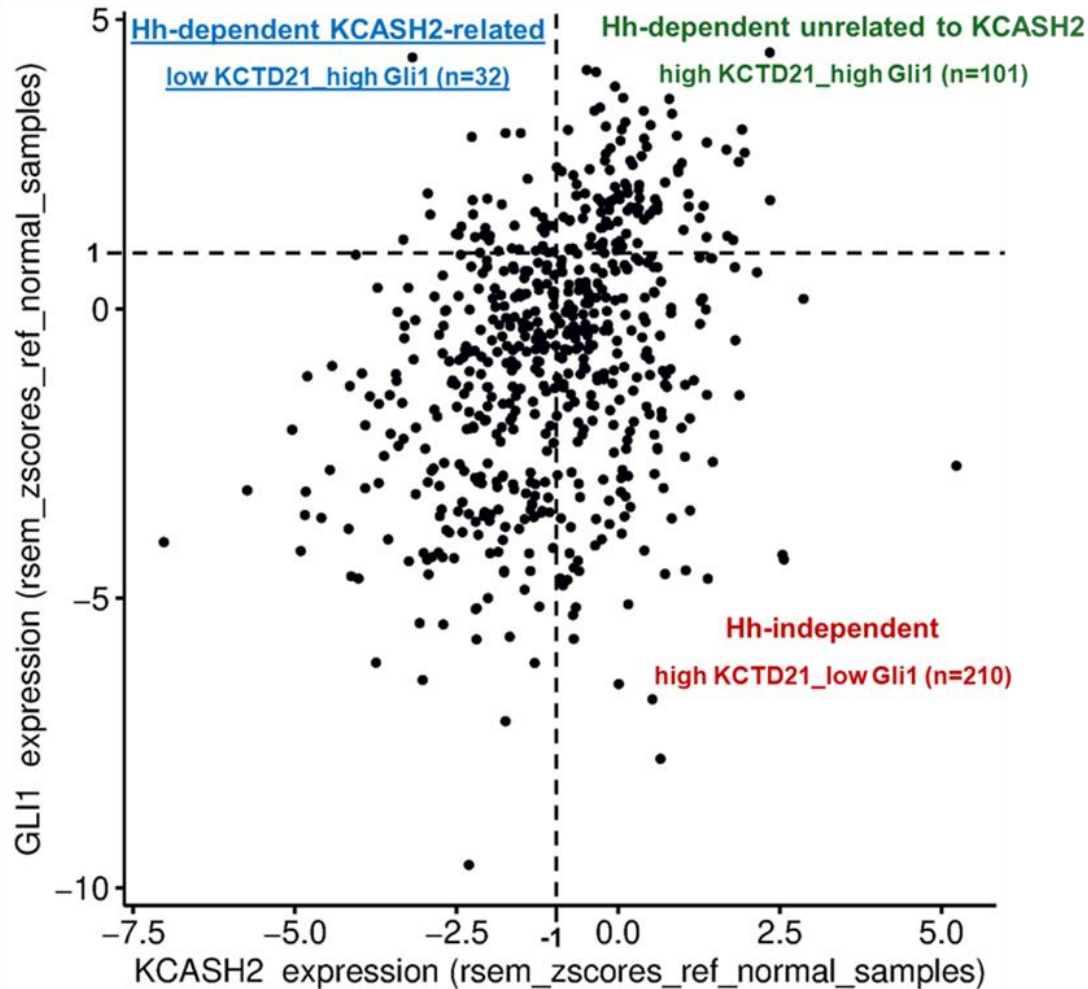
[Martin J Aryee](#)^{1,*}, [Andrew E Jaffe](#)^{2,3}, [Hector Corrada-Bravo](#)⁴, [Christine Ladd-Acosta](#)⁵, [Andrew P Feinberg](#)^{2,6},
[Kasper D Hansen](#)^{2,7,*}, [Rafael A Irizarry](#)^{2,*}

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PMCID: PMC4016708 PMID: [24478339](#)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4016708/>

Project 2



Study series: 592 CRC samples, 3 groups of interest

- **Hh-dependent KCASH2-related tumors (low KCASH2/high Gli1) N=32**
- Hh-independent tumors (high KCASH2/low Gli1) N=210
- Hh-dependent tumors unrelated to KCASH2 (low KCASH2/high Gli1) N=101

Input files:

- β -values (downloaded from cBioporta Colorectal Adenocarcinoma TCGA Pan-Cancer Atlas dataset)
- Meta data (sample information)

Differential methylation analysis (2 analyses):

Hh-dependent KCASH2-related tumors vs Hh-independent tumors

Hh-dependent KCASH2-related tumors vs Hh-dependent tumors unrelated to KCASH2

GO and pathways analysis

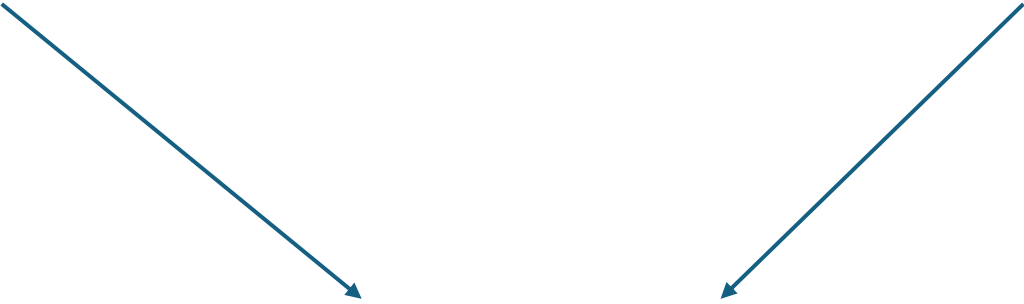
Summary

Project 1 (mouse)

- Input data: RNAseq raw counts
- Differential gene expression analysis in
 - KCASH2-KO tumors vs KCASH2-WT tumors
 - KCASH2-KO normals vs KCASH2-WT normals
 - KCASH2-KO tumors vs KCASH2-KO normals (paired)
 - KCASH2-WT tumors vs KCASH2-WT normals (paired)

Project 2 (human)

- Input data: β -values
- Gene-level methylation differences in
 - **Hh-dependent KCASH2-related tumors vs Hh-independent tumors**
 - **Hh-dependent KCASH2-related tumors vs Hh-dependent tumors unrelated to KCASH2**

- 
- Biological interpretation of differentially expressed/methylated genes (shared and unique across the different comparisons)